

Linkage and a possible Calabi–Yau smoothing

Questions for Marcos Jardim

Daniel · 29 September 2026

1. My example and the goal

I am studying the Stanley–Reisner scheme of the Grünbaum–Sreedharan triangulated 3-sphere on eight vertices. In the coordinates of my notes,

$$S = \mathbb{C}[x_1, \dots, x_8], \quad X_0 = \text{Proj}(S/I_0) \subset \mathbb{P}^7,$$

where the sixteen minimal generators are

$$\begin{aligned} I_0 = (& x_6x_7x_8, x_4x_6x_8, x_3x_7x_8, x_3x_5x_7, \\ & x_3x_4x_8, x_2x_7x_8, x_2x_5x_7, x_2x_5x_6, \\ & x_2x_4x_7, x_2x_4x_6, x_1x_4x_6, x_1x_4x_5, \\ & x_1x_3x_8, x_1x_3x_6, x_1x_3x_5, x_1x_2x_5). \end{aligned}$$

This is a reduced union of twenty coordinate $\mathbb{P}^3$'s, of degree 20 and codimension 4. Its coordinate ring is Gorenstein, with minimal graded free resolution

$$0 \longrightarrow S(-8) \longrightarrow S(-5)^{16} \longrightarrow S(-4)^{30} \longrightarrow S(-3)^{16} \longrightarrow S \longrightarrow S/I_0 \longrightarrow 0.$$

My hope is to smooth X_0 to a Calabi–Yau threefold. I believe I have found a smoothing through deformation computations, but I am still working through the proof, including flatness, algebraization and smoothness of the general fibre. I therefore regard this as work in progress.

Kapustka suggested studying the special fibre via linkage. The goal of this question is a *geometric linkage description of the resulting Calabi–Yau*: ideally, a construction from a simpler, recognizable variety. A rational starting variety would be especially appealing.

2. A motivating example of Kapustka and collaborators

In [1, §4.4], start with a linear $\Pi \cong \mathbb{P}^3 \subset \mathbb{P}^7$ and a general complete intersection $W = V(q_1, q_2, q_3)$ of three quadrics containing Π . Choose a hyperplane $H = V(h)$ through Π . The residual threefold D and a second residual threefold X are obtained by

$$\begin{aligned} I_D &= (q_1, q_2, q_3, h) : I_\Pi, \quad \deg D = 8 - 1 = 7, \\ I_X &= (q_1, q_2, q_3, c) : I_D, \quad \deg X = 24 - 7 = 17, \end{aligned}$$

where c is a suitable cubic containing D . Thus $W \cap H = \Pi \cup D$ and $W \cap V(c) = D \cup X$. Here $J : I = \{f : fI \subseteq J\}$ defines the residual scheme.

The authors obtain smooth arithmetically Gorenstein Calabi–Yau threefolds X ; Lemma 4.3 shows that the construction reaches a general member of their deformation family. This degree-17 example illustrates the kind of description I seek for my degree-20 problem.

3. How does a chosen link vary in a family?

Suppose, conditionally on the smoothing argument, that $\mathcal{X} \subset \mathbb{P}^7 \times B$ is a flat embedded family over a smooth pointed curve $(B, 0)$, with special fibre X_0 and smooth general fibre. Choose a codimension-four complete intersection Z_0 containing X_0 , and let Y_0 be its residual:

$$I_{Z_0} = (F_1, F_2, F_3, F_4) \subset I_0, \quad I_{Y_0} = I_{Z_0} : I_0.$$

Question 1. Under what hypotheses does this chosen link extend along the given deformation, after shrinking the base or an appropriate base change? More precisely, can we lift $X_0 \subset Z_0$ to $\mathcal{X} \subset \mathcal{Z} \subset \mathbb{P}^7 \times B$, with $\mathcal{Z}$ a relative complete intersection of the same multidegree, such that

$$\mathcal{I}_{\mathcal{Y}} = \mathcal{I}_{\mathcal{Z}} : \mathcal{I}_{\mathcal{X}}, \quad \mathcal{I}_{Y_b} = \mathcal{I}_{Z_b} : \mathcal{I}_{X_b} \quad \text{for every } b \in B,$$

and $\mathcal{Y} \rightarrow B$ is flat? I would like to understand the precise conditions ensuring both flatness and compatibility of residual formation with taking fibres.

In moduli language, fix the Hilbert polynomials and the multidegree $\mathbf{d}$ of Z . Let $\mathcal{F}_{\mathbf{d}}$ be the locus in the flag Hilbert scheme parametrizing such pairs $X \subset Z$. The relevant map is

$$p : \mathcal{F}_{\mathbf{d}} \longrightarrow \text{Hilb}(\mathbb{P}^7), \quad (X \subset Z) \longmapsto [X].$$

Is p smooth at a suitable $(X_0 \subset Z_0)$ in this example? For instance, if the F_i are cubics, does the ACM vanishing $H^1(\mathbb{P}^7, \mathcal{I}_{X_0}(3)) = 0$ supply the needed lifting of the four equations, leaving relative liaison to control the residual family?

4. Can the special fibre lead to a useful description?

Question 2. Does it make sense to choose a particularly understandable link, or sequence of links, on X_0 and carry it to the smooth fibres? The crucial issue seems to be how to choose the link so that the *general residual fibre* still has recognizable geometry.

More concretely, suppose I specify a geometric family $\mathcal{T}$ of candidate residual threefolds. Consider the space of linkage data (X, Y, Z) with $Y \in \mathcal{T}$. How would one test whether its projection to $\text{Hilb}(\mathbb{P}^7)$ dominates the component containing the proposed smoothing branch? Which tangent-space or obstruction calculation would be a useful first test? This would distinguish a construction of general smooth members from a special-fibre construction alone.

A connection with your questions. Your problem presentation on curves, sections of a common bundle and their quotient sheaves made me wonder whether the linkage should be encoded by a sheaf presentation that can be deformed. Your work on components and smoothable boundary sheaves, for example [2], suggests this perspective. In this codimension-four example, is there a useful sheaf or resolution datum to track for that purpose? I would appreciate a criterion or a comparable example indicating where to start.

References

- [1] S. Coughlan, Ł. Gołębowski, G. Kapustka and M. Kapustka, *Arithmetically Gorenstein Calabi–Yau threefolds in $\mathbb{P}^7$* , Electron. Res. Announc. Math. Sci. **23** (2016), 52–68, §4.4 and Lemma 4.3. [arXiv:1609.01195](https://arxiv.org/abs/1609.01195).
- [2] C. Almeida, M. Jardim and A. S. Tikhomirov, *Irreducible components of the moduli space of rank 2 sheaves of odd determinant on the projective space*, Adv. Math. **402** (2022), 108363. [arXiv:1903.00292](https://arxiv.org/abs/1903.00292).